

Prior : $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$

Likelihood : $x | \theta \sim \mathcal{N}(\theta, \sigma^2)$, known Variance.

$$P(\theta | x) = \frac{P(x | \theta) P(\theta)}{P(x)}$$

$$\propto \frac{1}{\sqrt{2\pi\sigma^2}} \left(\exp \left(- \left(\frac{(x-\theta)^2}{2\sigma^2} + \frac{(\theta-\mu_0)^2}{2\sigma_0^2} \right) \right) \right)$$

with precision substitute $\tau = \frac{1}{\sigma^2}$ $\tau_0 = \frac{1}{\sigma_0^2}$:

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(- \frac{1}{2} \left(\tau (x-\theta)^2 + \tau_0 (\theta-\mu_0)^2 \right) \right)$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(- \frac{1}{2} \left(\tau x^2 + \tau \theta^2 - 2\tau \theta x + \tau_0 \theta^2 + \tau_0 \mu_0^2 - 2\tau_0 \theta \mu_0 \right) \right)$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(- \frac{1}{2} \left((\tau + \tau_0) \theta^2 - 2(\tau x + \tau_0 \mu_0) \theta + C \right) \right)$$

Exp. terms : let $\tau_{post} = \tau + \tau_0$

$$= \tau_{post} \left[\theta^2 - 2 \frac{\tau x + \tau_0 \mu_0}{\tau_{post}} \theta \right] + C$$

$$\text{let } m = \frac{\tau_0 \mu_0 + \tau \kappa}{\tau_0 + \tau} :$$

$$= \tau_{\text{post}} [\theta^2 - 2m\theta] + C$$

$$= \tau_{\text{post}} [(\theta - m)^2 - m^2] + C$$

$$= \tau_{\text{post}} (\theta - m)^2 + C - \tau_{\text{post}} m^2$$

$$\text{So exp reduces to } -\frac{1}{2} \tau_{\text{post}} (\theta - m)^2 :$$

$$P(\theta/\kappa) \propto \exp\left(-\frac{1}{2} \tau_{\text{post}} (\theta - m)^2\right)$$

$$N(\mu_{\text{post}}, \sigma_{\text{post}}^2) \propto \exp\left(-\frac{(\theta - \mu_{\text{post}})^2}{2\sigma_{\text{post}}^2}\right)$$

$$\mu_{\text{post}} = m = \frac{\tau_0 \mu_0 + \tau \kappa}{\tau_0 + \tau} = \frac{\sigma^2 \mu_0 + \sigma_0^2 \kappa}{\sigma^2 + \sigma_0^2}$$

$$\sigma_{\text{post}}^2 = \frac{1}{\tau_{\text{post}}} = \frac{\sigma_0^2 \sigma^2}{\sigma_0^2 + \sigma^2}$$

or

$$\frac{1}{\sigma^2_{post}} = \frac{1}{\sigma^2_0} + \frac{1}{\sigma^2}$$

Matching Coef Method

$$P(\theta|x) \propto P(x|\theta) \cdot P(\theta)$$

Conditioning on hyperparameters / Explicit Parametrisation:

$$P(\theta|x, \theta_0, \sigma_0^2, \sigma^2) \propto \underbrace{P(x|\theta, \sigma^2)}_{\text{likelihood}} \underbrace{P(\theta|\theta_0, \sigma_0^2)}_{\text{Prior}}$$

Parametric / Conjugate Model formulation:

$$\begin{aligned} N(\theta|\theta_{\text{post}}, \sigma_{\text{post}}^2) &\propto N(x|\theta, \sigma^2) N(\theta|\theta_0, \sigma_0^2) \\ &\propto \exp\left(-\frac{1}{2}\left(\frac{(x-\theta)^2}{\sigma^2} + \frac{(\theta-\theta_0)^2}{\sigma_0^2}\right)\right) \end{aligned}$$

$$\text{let } \tau = \frac{1}{\sigma^2} \quad \tau_0 = \frac{1}{\sigma_0^2}$$

$$\text{Exponent} = -\frac{1}{2} \left[\tau (x^2 + \theta^2 - 2x\theta) + \tau_0 (\theta^2 + \theta_0^2 - 2\theta\theta_0) \right]$$

$$= -\frac{1}{2} \left((\tau + \tau_0) \theta^2 - (2x\tau + 2\theta_0\tau_0) \theta + c \right)$$

Inside Bracket $\propto (\tau_0 + \tau) \theta^2 - (2x\tau + 2\theta_0\tau_0) \theta$

$$\begin{aligned} \text{Posterior inside bracket} &= \tau_{\text{post}} (\theta - \theta_{\text{post}})^2 \\ &= \tau_{\text{post}} (\theta^2 + \theta_{\text{post}}^2 - 2\theta\theta_{\text{post}}) \end{aligned}$$

$$\underbrace{\tau_{\text{post}} = \tau_0 + \tau}_{\text{Precisions add}}$$

$$\underbrace{\theta_{\text{post}} = \frac{x\tau + \theta_0\tau_0}{\tau_{\text{post}}}}_{\text{Precision weighted average.}}$$